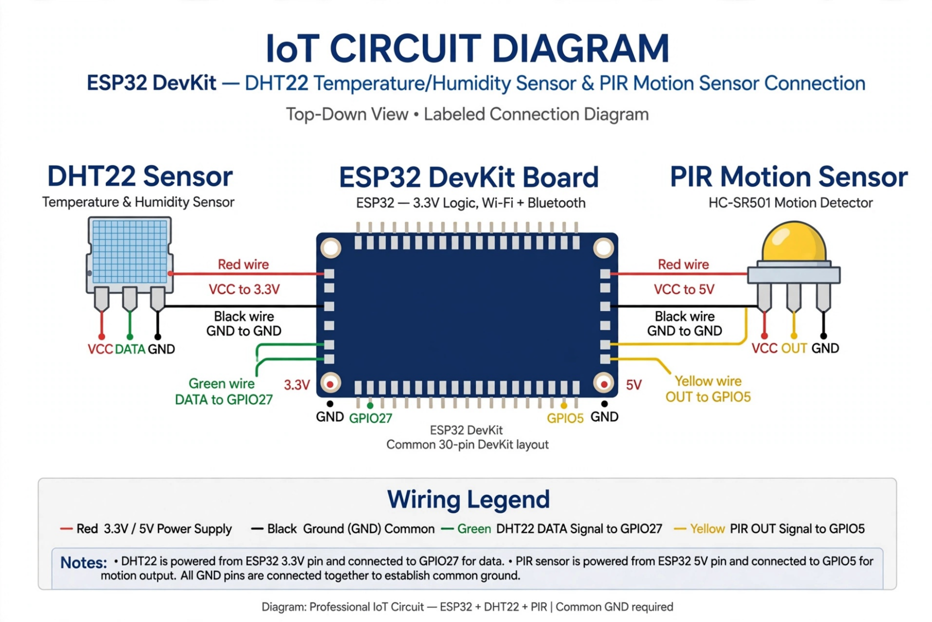
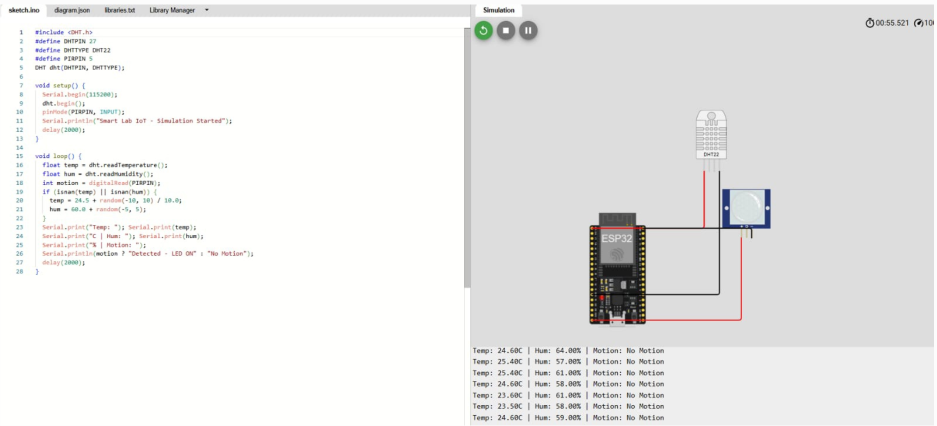

IoT CIRCUIT DIAGRAM
ESP32 DevKit — DHT22 Temperature/Humidity Sensor & PIR Motion Sensor Connection
Top-Down View • Labeled Connection Diagram
DHT22 Sensor
Temperature & Humidity Sensor
ESP32 DevKit Board
ESP32 — 3.3V Logic, Wi-Fi + Bluetooth
PIR Motion Sensor
HC-SR501 Motion Detector
Red wire
VCC to 3.3V
Black wire
GND to GND
VCC DATA GND
Green wire
DATA to GPIO27
3.3V
GND GPIO27
Red wire
VCC to 5V
Black wire
GND to GND
VCC OUT GND
5V
Yellow wire
OUT to GPIO5
GPIO5 GND
ESP32 DevKit
Common 30-pin DevKit layout
Wiring Legend
— Red 3.3V / 5V Power Supply — Black Ground (GND) Common — Green DHT22 DATA Signal to GPIO27 — Yellow PIR OUT Signal to GPIO5
Notes: • DHT22 is powered from ESP32 3.3V pin and connected to GPIO27 for data. • PIR sensor is powered from ESP32 5V pin and connected to GPIO5 for motion output. All GND pins are connected together to establish common ground.
Diagram: Professional IoT Circuit — ESP32 + DHT22 + PIR | Common GND required


sketch.ino
diagram.json
libraries.txt
Library Manager
#include <DHT.h>
#define DHTPIN 27
#define DHTTYPE DHT22
#define PIRPIN 5
DHT dht(DHTPIN, DHTTYPE);
void setup() {
Serial.begin(115200);
dht.begin();
pinMode(PIRPIN, INPUT);
Serial.println("Smart Lab IoT - Simulation Started");
delay(2000);
}
void loop() {
float temp = dht.readTemperature();
float hum = dht.readHumidity();
int motion = digitalRead(PIRPIN);
if (isnan(temp) || isnan(hum)) {
temp = 24.5 + random(-10, 10) / 10.0;
hum = 60.0 + random(-5, 5);
}
Serial.print("Temp: "); Serial.print(temp);
Serial.print("C | Hum: "); Serial.print(hum);
Serial.print("% | Motion: ");
Serial.println(motion ? "Detected - LED ON" : "No Motion");
delay(2000);
}
Simulation
00:55.521
DHT22
ESP32
Temp: 24.60C | Hum: 64.00% | Motion: No Motion
Temp: 25.40C | Hum: 57.00% | Motion: No Motion
Temp: 25.40C | Hum: 61.00% | Motion: No Motion
Temp: 24.60C | Hum: 58.00% | Motion: No Motion
Temp: 23.60C | Hum: 61.00% | Motion: No Motion
Temp: 23.50C | Hum: 58.00% | Motion: No Motion
Temp: 24.60C | Hum: 59.00% | Motion: No Motion